TermGraph: A Certified Glassbox Exchange Architecture for GBM-Class Models

Patricio Alejandro Mery Caro
Independent Researcher, La Serena, Chile

Abstract

Gradient-boosted trees are still the default for tabular prediction. Their explanations are post hoc, and the deployed artifact cannot be edited. Explainable boosting machines (EBMs) and neural additive models (NAMs) are glassboxes, but both are fixed-order learners; neither acts as an exchange layer for an existing GBM. TermGraph compiles a GBM into a sum of low-order table terms. The term ledger is exact, feature values use a declared sharing convention, terms can be edited after training, and a controller chooses between compact and max-accuracy artifacts. On the KIRK planetary-stability task, a 22-term TermGraph compiled from an EBM plus an order-3 residual reaches 0.9427 ± 0.0048 AUC over five seeds, against 0.9417 ± 0.0049 for EBM GA^{2}M and 0.9206 for NAM. A five-term compact artifact reaches 0.9296 at the same budget, where EBM reaches 0.9252. On a controlled 3-way benchmark, TermGraph reaches 0.8297 against 0.6989 for pairwise EBM. The certified 22-term table matches its source to 5.3×10^{-15} maximum logit error, and a fused runtime reaches 8.72× on eight threads (5.66M rows/s) with a 0.66 MB binary artifact. I also report where EBM remains the better choice and where a marginal ANOVA diagnostic gives a misleading picture.

Keywords: glassbox models, explainable artificial intelligence, explainable boosting machines, SHAP, gradient boosting, tabular data

1 Introduction

Tabular prediction is still dominated by gradient-boosted decision trees (GBMs) such as XGBoost, LightGBM, and CatBoost (Chen and Guestrin, 2016; Ke et al., 2017; Prokhorenkova et al., 2018). They are accurate and robust, and they are easy to tune. The deployed object, however, is an ensemble of trees whose decisions cannot be inspected directly. SHAP and TreeSHAP (Lundberg and Lee, 2017; Lundberg et al., 2020) give post-hoc attributions, but the explanation remains a separate object: it cannot be edited, it depends on a baseline, and it does not give a replacement model.

Additive glassboxes take a different route. Generalized additive models (GAMs) (Hastie and Tibshirani, 1990), explainable boosting machines (EBMs/GA^{2}M) (Lou et al., 2012, 2013), and neural additive models (NAMs) (Agarwal et al., 2021) describe the prediction as a sum of low-order functions. This makes the model readable, but the hypothesis class is fixed in advance. EBM stops at pairwise interactions; NAM stops at univariate shapes. Neither can take an existing GBM and return a certified, editable artifact.

I ask whether a glassbox can serve as that exchange layer. I test the question on KIRK, a planetary-stability classifier, and on a matched synthetic benchmark. Both tests support the role. A 22-term TermGraph matches the EBM plus order-3 hybrid, and a five-term

artifact is competitive at the same term budget. EBM remains the better low-order model, which is a useful boundary rather than a defect.

A term graph represents the model as a sum of low-order tables. A compiler turns EBM terms and a constrained XGBoost term into those tables, and a Glassbox Contract plus a budget controller decide which artifact can be deployed. The empirical section reports the KIRK comparison, a controlled 3-way test, post-hoc term edits, and the runtime. I keep the negative results in the paper because they mark where the claim stops.

2 Related Work

Research on additive glassboxes begins with GAMs (Hastie and Tibshirani, 1990), which model a response as a sum of univariate shape functions. EBM and GA^{2}M (Lou et al., 2012, 2013) add pairwise terms with boosted stumps; InterpretML provides a practical implementation (Nori et al., 2019). NAM (Agarwal et al., 2021) replaces the shape functions with small neural networks. These are the closest baselines, and they are the reason the paper reports where TermGraph wins instead of making a blanket claim.

Post-hoc attribution follows a different tradition. SHAP (Lundberg and Lee, 2017) is the standard attribution framework, and TreeSHAP (Lundberg et al., 2020) is exact for tree ensembles. The limitation is structural: a post-hoc explanation is not a model artifact and cannot be edited. Rudin (Rudin, 2019) argues for inherently interpretable models when stakes are high. TermGraph follows that direction and adds an exchange and certificate layer.

XGBoost, LightGBM, and CatBoost (Chen and Guestrin, 2016; Ke et al., 2017; Prokhorenkova et al., 2018) define the accuracy envelope for tabular learners. TermGraph uses XGBoost for the order-3 term and EBM for the low-order terms. The deployed object is a table graph, not a tree ensemble.

3 Architecture

3.1 Term graph and blocks

A TermGraph is

$$f(x) = b + \sum_{j=1}^{K} g_j(x_{S_j}), \qquad |S_j| \le d, \tag{1}$$

where $S_j \subseteq [p]$ is the support of term j, $g_j : \mathbb{R}^{|S_j|} \to \mathbb{R}$ is a centered term, and b is the bias. Supports may overlap; the graph is a hypergraph. Complexity is the vector

$$C(G) = (K,\, d,\, \text{overlap},\, \text{cells}), \qquad \text{overlap} = \max_i |\{j : i \in S_j\}|. \tag{2}$$

The implementation exposes a reusable block specification (`TermBlock`: inputs, basis, learner, composition, order) and a `ModelSpec` for candidate artifacts. The compact artifact is a partition of the 11 features; the max-accuracy artifact is a declared hypergraph.

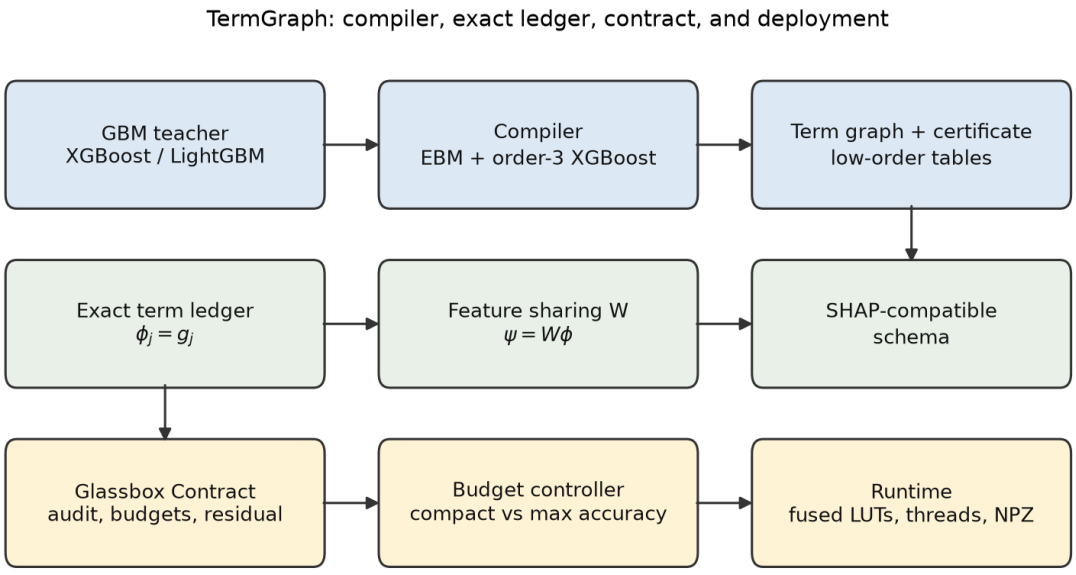


Figure 1: TermGraph compiles a GBM into a low-order term graph. The exact term ledger, declared feature sharing, Glassbox Contract, controller, and fused deployment runtime are separate layers.

3.2 Glassbox Contract

A deployed graph is admissible only if

$$K \le K_{\max}, \quad d \le d_{\max}, \quad \text{cells} \le C_{\max}, \quad \text{additivity} \le \varepsilon, \quad \text{certificate present.} \tag{3}$$

The contract also requires a rendering protocol: a 1D curve, a 2D heatmap, or 3D slices. Cell count alone is not readability. The controller refuses a budget that no admissible artifact can meet.

4 Mathematical Core

4.1 Table terms

For term j and input l, let $e_{j,l,0} < \cdots < e_{j,l,m_{j,l}}$ be explicit edges, with sentinel outer edges. The bin index is

$$\kappa_{j,l}(x_l) = \text{clip}\left(\text{searchsorted}(e_{j,l}, x_l) - 1,\ 0,\ m_{j,l} - 1\right), \tag{4}$$

and the term is

$$g_j(x_{S_j}) = \theta_j[\kappa_{j,1}(x_{i_1}), \ldots, \kappa_{j,|S_j|}(x_{i_{|S_j|}})] - c_j, \tag{5}$$

where θ_j is the value table and c_j is the centering constant. A table compiler is exact iff the source term is constant on every cell of the Cartesian product of its edges.

4.2 Two attribution levels

The *term ledger* is

$$\phi_j(x) = g_j(x_{S_j}), \qquad f(x) = b + \sum_j \phi_j(x), \tag{6}$$

which is exact by construction. Feature attribution uses a sharing matrix $W \in \mathbb{R}^{p\times K}$:

$$W_{i,j} \geq 0, \quad W_{i,j} = 0 \text{ if } i \notin S_j, \quad \sum_i W_{i,j} = 1, \quad \psi_i(x) = \sum_j W_{i,j}\phi_j(x). \tag{7}$$

The default is equal split, $W_{i,j} = 1/|S_j|$. This is the Shapley value of the within-term symmetric game, *not* the full-model Shapley value when terms overlap. If full-model Shapley is required, it must be computed separately; the ledger is a coarsening.

4.3 Two order definitions

Under a product baseline $P_\perp$, the functional-ANOVA tail energy is

$$E_{>d} = \sum_{|A|>d} \mathrm{Var}_{P_\perp}(f_A). \tag{8}$$

This bounds approximation under $P_\perp$, not under the data distribution P. The effective order is

$$d_{\text{eff}}(P, \delta) = \min\{d : R_d(P) - R^*(P) \leq \delta\}. \tag{9}$$

The two can disagree sharply: on KIRK, marginal ANOVA assigns only 43.3% of the energy to order ≤ 3, while an order-3 glassbox captures 93.5% in distribution. The paper reports both and does not use the marginal number as the deployment limit.

4.4 Exactness

Table 1 separates the exact claims from the conventions. The term ledger is exact by construction; feature values use a declared sharing matrix; the marginal ANOVA tail is a product-baseline diagnostic, not a data-distribution guarantee.

5 KIRK Case Study

5.1 Data and protocol

KIRK classifies the long-term stability of compact planetary systems from a short integration (Tamayo et al., 2020; Thadhani et al., 2025; Hussain and Tamayo, 2020; Rein and Tamayo, 2017). The cache contains 12,000 real initial conditions with 11 SPOCK 2.0 features; 9,767 are short-stable and used for training and evaluation. The teacher is the SPOCK 2.0 XGBoost classifier (Thadhani et al., 2025). The target is the real 10^9-orbit label. I use five seeds, an 85/15 split, and exclude short-unstable systems from scoring, following the KIRK protocol.

The max-accuracy artifact uses an EBM GA^{2}M with 10 pairwise interactions for the low-order terms and one constrained XGBoost order-3 residual term on `EMcrossnear`, `MMRstrengthnear`, and `EMfracstdnear`. The compact artifact uses a partition of five terms: one 3-way block and four pairs.

Claim	Exact?	Caveat
$f = b + \sum_j \phi_j$	yes	by construction
table compiler	yes iff constant on cells	otherwise δ
term ledger ϕ_j	yes	baseline P declared
feature attribution ψ_i	convention	sharing matrix W
full-model Shapley	separate computation	not the ledger when overlapping
marginal ANOVA tail $E_{>d}$	yes under $P_\perp$	not the P limit

Table 1: Exactness claims in this paper, separated by object. The term ledger is exact by construction. Feature attribution is a declared sharing convention, not full-model Shapley. The marginal ANOVA tail is a product-baseline diagnostic and does not bound the data-distribution risk.

Model	Terms	AUC truth	Log-loss	Accuracy
NAM 12k	11	0.9206 ± 0.0068	–	0.8409
EBM GAM	11	0.9386 ± 0.0054	0.2992	–
EBM GA^{2}M	21	0.9417 ± 0.0049	0.2917	0.8718
TermGraph max (EBM + order 3)	22	$\mathbf{0.9427 \pm 0.0048}$	**0.2895**	**0.8729**
TermGraph compact	5	0.9296	0.3200	0.8615
EBM at 5-term budget	5	0.9252	–	–
Unconstrained XGBoost	–	0.9381	–	–
SPOCK teacher	–	0.9566	–	–

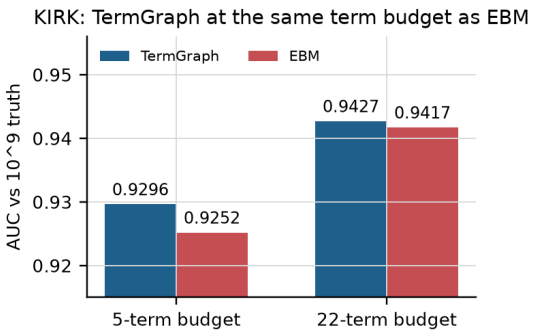
Table 2: KIRK results. AUC is against the real 10^9-orbit labels and higher is better; log-loss is lower. The max and EBM GA^{2}M rows are five-seed means. The compact row is the deployed five-term artifact on its certified fold. EBM at the five-term budget uses five features because EBM needs one term per feature. The unconstrained XGBoost row is the clean internal-validation reference, and the teacher is the 100k SPOCK model.

5.2 Baselines and metrics

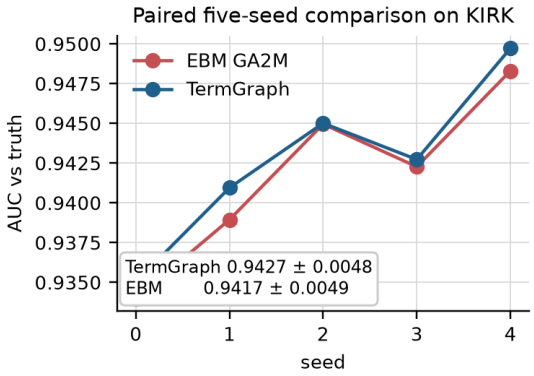
Baselines are NAM, EBM GAM, EBM GA^{2}M, an unconstrained XGBoost student trained on the same data, and the SPOCK teacher. The primary metric is AUC against the real labels; I also report log-loss and accuracy.

5.3 Results

Table 2 reports the KIRK comparison; the max and EBM GA^{2}M rows are five-seed means. The max model reaches 0.9427 ± 0.0048 AUC over five seeds, against 0.9417 ± 0.0049 for



(a) AUC at five and 22 terms. The five-term TermGraph uses all 11 features; the five-term EBM uses five.



(b) Paired seeds. Each line is one 85/15 split; both models use the same split.

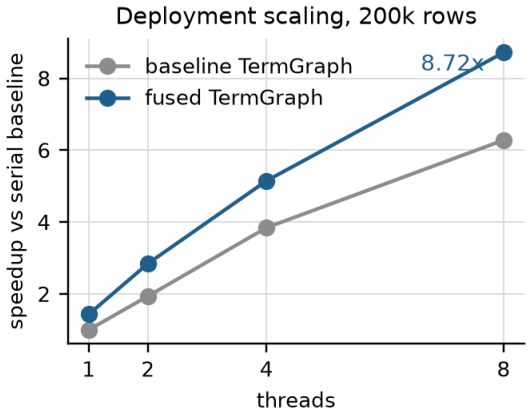
Figure 2: KIRK performance. The compact artifact is competitive at a five-term budget, and the max artifact matches the EBM plus order-3 hybrid on the same five seeds.

EBM GA^{2}M and 0.9206 for NAM. The paired gain over EBM is +0.0010, and it is positive on all five seeds. The compact artifact reaches 0.9296 at a five-term budget, while EBM reaches 0.9252 because it cannot merge features into multi-feature terms.

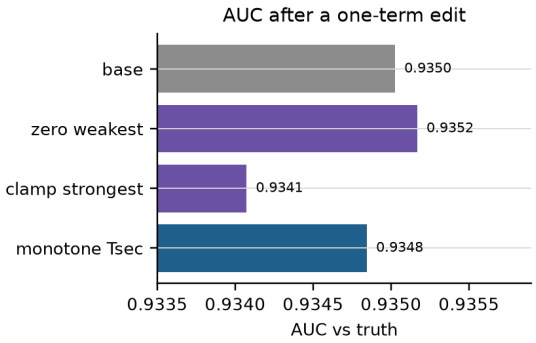
Figure 2 shows the paired seeds and the fixed-budget comparison. The comparison is mixed. On raw KIRK AUC, EBM GA^{2}M (0.9417) beats both the deployed compact artifact (0.9296) and the earlier full-capacity five-term experiment (0.9364). The defensible claim is narrower. TermGraph adds order 3 and keeps a certified, editable ledger, and it does so without losing the low-order baseline.

5.4 Compilation and hardware

Figure 3 reports the deployment and edit measurements. The EBM terms are compiled with EBM's own cut points; the 3-way term uses the union of its tree thresholds. On the certified fold, the table matches the EBM + order-3 hybrid to 5.3×10^{-15} maximum logit error, and additivity is 0.0. A fused runtime bins each unique feature/edge pair once per batch, flattens term tables, and accumulates without materializing the (n, K) matrix. On 200,000 rows it reaches 8.72× on eight threads (5.66M rows/s) with a maximum difference of 5.3×10^{-15}. The binary NPZ artifact is 0.66 MB versus 6.0 MB JSON and loads 7.4× faster.



(a) Speedup against the serial baseline on 200k rows. The fused runtime bins each unique feature and edge set once per batch.



(b) AUC before and after zeroing, clamping, or monotonizing one term. The remaining-term maximum delta is 0.0.

Figure 3: Fused table inference and post-hoc term surgery. The edits change one term and leave the other term contributions unchanged; log-loss improves for the clamp and monotone edits.

5.5 Term surgery and the controller

Zeroing the weakest term changes only that term; the remaining-term maximum delta is 0.0. Clamping the strongest term and enforcing monotonicity in `Tsec` also leave the rest unchanged; the monotone edit reduces 1,365 violations to zero while improving log-loss. The controller selects the compact artifact at budgets 5–6 and the max artifact at 22; budget 4 returns no admissible model.

5.6 Data scaling

Figure 4 reports the scaling record. It cautions against reading the five-seed result as a 100k-data claim. From 1k to 8.3k truth-labelled systems, TermGraph improves from 0.9152 to 0.9349 AUC; distillation of the teacher plateaus around 0.9264; the unconstrained GBM improves from 0.9104 to 0.9384. Truth labels scale, distillation saturates, and the full 100k archive was not trained here.

6 Controlled 3-way Test

Figure 5 reports the controlled test. I generate three synthetic truths: additive, pairwise, and 3-way (within-block 3-way terms, no cross-block pairs). The matched TermGraph uses the corresponding blocks; EBM uses 10 pairwise interactions; the unconstrained GBM is the reference.

On the 3-way truth, TermGraph reaches 0.8297 ± 0.0034 AUC, the unconstrained GBM 0.8217 ± 0.0018, and EBM GA^{2}M 0.6989 ± 0.0022. On the pairwise truth, TermGraph and EBM are statistically comparable (0.8304 vs 0.8319), and both beat the unconstrained

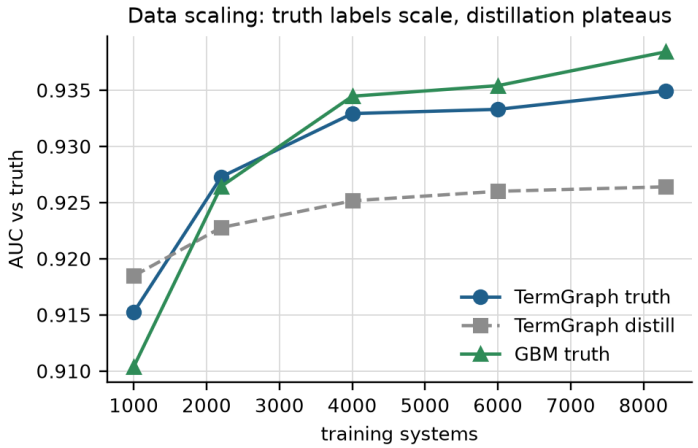


Figure 4: Data scaling. The x-axis is the number of truth-labelled training systems, and the y-axis is AUC against truth. Truth-label training keeps improving; teacher distillation plateaus. The full 100k archive is out of scope.

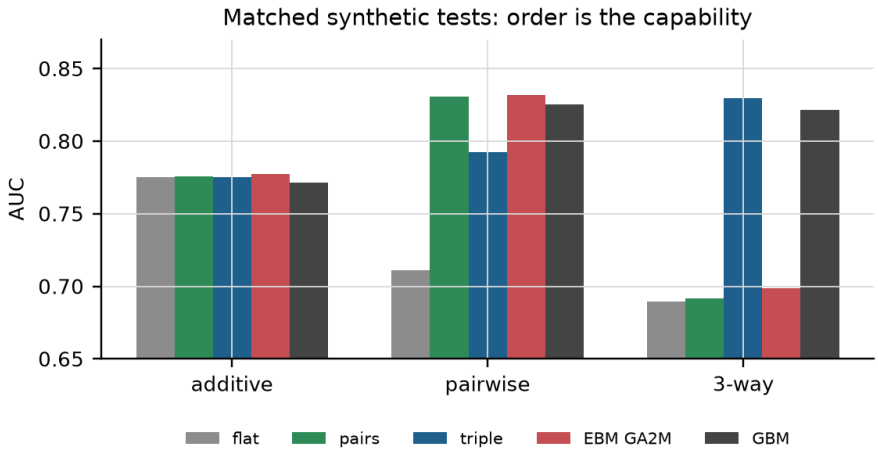


Figure 5: Matched synthetic tests. Each group is a known truth: additive, pairwise, or 3-way. Bars are test AUC over three seeds. On the 3-way truth, TermGraph (triple) reaches 0.8297 versus 0.6989 for pairwise EBM and 0.8217 for the unconstrained GBM. EBM cannot represent the signal by construction.

GBM (0.8255). On the additive truth, EBM is slightly ahead (0.7779 vs 0.7752). Order 3 is a capability advantage here, not a blanket accuracy advantage.

7 Discussion

The exact part of TermGraph is the term ledger. Every prediction is a bias plus a sum of terms, so the ledger adds up by construction. The table compiler is exact when the source term is constant on the cells of its edges, and the certified KIRK fold meets that condition to floating-point noise. Feature values are a different object: they use the declared sharing matrix W, and they are not full-model Shapley when terms overlap. The controller is a budget selector over expected performance; it does not provide a non-inferiority guarantee.

EBM remains the stronger low-order model. On KIRK it beats the compact five-term artifact on raw AUC, and it is the right choice when the effective order is at most two. TermGraph is useful when the data needs order 3 and when the artifact must stay editable. This is why the paper reports both the max and compact artifacts rather than a single winner.

The main structural caveat is identifiability. Overlapping terms are not unique, so the max artifact's ledger is exact for the declared EBM term set but is not a canonical functional-ANOVA decomposition. The compact artifact is a partition, which makes it the cleaner formal object.

The study covers one real task, KIRK, with external data and an external teacher. I do not include a human user study, categorical or missing-value benchmarks, multi-class objectives, or training on the full 100k archive. The 3D term is inspected through slices; cell count alone is not readability.

The runtime is a sum of lookup tables. There are no dense matrix multiplications and no multiplications, and the terms are independent. The fused kernel and the binary NPZ artifact are the deployment path evaluated here.

8 Conclusion

TermGraph turns a GBM-class model into a certified sum of low-order terms. On KIRK it matches the EBM plus order-3 hybrid to floating-point noise and reaches 0.9427 AUC against 0.9417 for EBM and 0.9206 for NAM. A five-term artifact reaches 0.9296 at a five-term budget, and the controlled 3-way test gives 0.8297 against 0.6989 for EBM. The same artifact runs at 5.66M rows/s on eight threads and ships as a 0.66 MB NPZ file.

The evidence is limited to KIRK and the controlled benchmark. EBM remains the better low-order model. Within that boundary, TermGraph carries a GBM into a table graph with an exact ledger and editable terms, which the fixed-order glassboxes do not offer.

Data and code availability

The code, paper source, figures, and compiled artifacts are available in the TermGraph repository, `github.com/AleeeeM2008/termgraph`. The KIRK data and the SPOCK teacher are external resources; the repository includes the derived caches, certificates, and artifacts used in this paper.

Competing interests

The author declares no competing interests.

Disclosure

This code was assisted by DeepSeek V4.1 Flash to speed up investigation. The architecture, logic, experiment design, and verification were human-designed and reviewed; the assistant was used for implementation, plotting, and manuscript drafting. All reported results are reproducible from the included artifacts and scripts.

References

T. Chen and C. Guestrin. XGBoost: A scalable tree boosting system. KDD, 2016.

G. Ke et al. LightGBM: A highly efficient gradient boosting decision tree. NeurIPS, 2017.

L. Prokhorenkova et al. CatBoost: Unbiased boosting with categorical features. NeurIPS, 2018.

S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. NeurIPS, 2017.

S. M. Lundberg et al. From local explanations to global understanding with explainable AI for trees. Nature Machine Intelligence, 2020.

T. Hastie and R. Tibshirani. Generalized Additive Models. Chapman & Hall, 1990.

Y. Lou, R. Caruana, and J. Gehrke. Intelligible models for classification and regression. KDD, 2012.

Y. Lou, R. Caruana, J. Gehrke, and G. Hooker. Accurate intelligible models with pairwise interactions. KDD, 2013.

R. Agarwal et al. Neural additive models: Interpretable machine learning with neural nets. NeurIPS, 2021.

H. Nori et al. InterpretML: A unified framework for machine learning interpretability. arXiv:1909.09223, 2019.

C. Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence, 2019.

D. Tamayo et al. Predicting the stability of planetary systems with deep learning. PNAS, 2020.

J. Thadhani, K. Ba, H. Rein, and D. Tamayo. SPOCK 2.0. RNAAS, 2025.

N. Hussain and D. Tamayo. The effects of chaos on planetary stability. MNRAS, 2020.

H. Rein and D. Tamayo. REBOUND: An open-source multi-purpose N-body code. MNRAS, 2017.